01 / Chapter map and relevant tutorial questions

BBMS2006-4-Applied pulmonary physiology-25Sept2026.pdf | 12 source pages

Two environmental changes, different mechanisms

Altitude: barometric pressure ↓ → inspired PO_2 ↓ → hypoxic ventilatory drive and acclimatization.
Diving: ambient pressure ↑ → gas partial pressures / dissolved gas ↑; ascent also expands trapped gas.

Tutorial question	Relationship to this lecture	Notes page
Q4 | Altitude: hyperventilation and acidosis	Direct: hypoxic response versus respiratory alkalosis.	3
Q7 | Hyperventilation at unchanged inspired O_2	Application: alveolar ventilation can improve PAO_2.	2
Q12 | CA inhibitor and HCO_3^- excretion	Cross-topic: pharmacology supports the renal acclimatization mechanism.	3
Q13 | Diver: nitrogen fraction unchanged	Direct: total pressure, PN_2 and dissolved nitrogen.	7
Q14 | Equal minute ventilation	Direct: dead-space volume and effective alveolar ventilation.	5
Q3, Q5, Q6 | Content, affinity and hypoxia	Supporting Lecture 2 concepts; used here without repeating all three original answers.	2–4
Q11 | Chronic hypercapnia	Lecture 3; adaptation is related but this is not the altitude stimulus.	—
Q1, Q2, Q8, Q9, Q10	Mainly other lectures: mechanics, regional V/Q or CO poisoning.	—

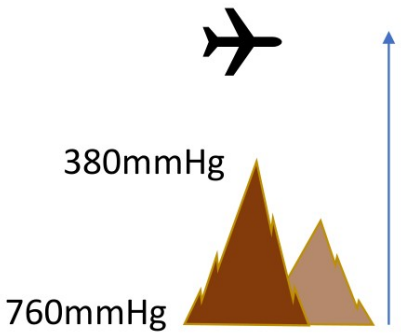
Read the appropriate law for the question

Law	Quantity being explained	Main use here
Dalton	Partial pressure = fraction × total pressure.	Altitude PO_2 and diving PN_2.
Henry	Equilibrium dissolved concentration = α × partial pressure.	Dissolved gases in blood/tissue.
Boyle	$P_1V_1 = P_2V_2$ for a fixed gas amount at constant temperature.	Expansion of trapped gas during ascent.

Tutorial Q# preserves original wording/numbering from BBMS 2006 Tutorial.pdf, p. 1; answers are reasoned suggestions, not an official key. Original diagrams and visible annotations are retained for comparison, not automatically accepted as correct. Answer-check labels identify reviewed choices; CORRECTION flags misleading source claims. [Extension] marks supporting material beyond the slides. References use actual lecture PDF pages.

02 / Altitude: lower pressure, not less O_2 fraction

Lecture PDF p. 4 | Tutorial Q7 | Quantitative alveolar gas calculation is an extension



Original mountain/pressure illustration, lecture p. 4.

Primary causal chain

PB ↓ while FiO_2 stays about 0.21 → inspired PIO_2 ↓ → alveolar PAO_2 ↓ → arterial PaO_2 ↓ → dissolved O_2 ↓ and Hb saturation can fall → arterial O_2 content ↓.
This is hypobaric hypoxia: reduced pressure, not a lower oxygen percentage in ordinary air.

Symbol / relationship	Meaning / condition
PB; FiO_2	Barometric pressure; fraction of inspired oxygen.
Dry inspired $PO_2 = FiO_2 \times PB$	At 760 mmHg: 159.6; at 380 mmHg: 79.8 mmHg.
$PIO_2 = FiO_2 \times (PB - 47)$	Humidified inspired gas at 37°C; 47 mmHg is water-vapour pressure.
$PAO_2 \approx PIO_2 - PaCO_2/R$	Simplified alveolar gas equation; R ~0.8. PA = alveolar, Pa = arterial.

Tutorial Q7 · Original tutorial question

Hyperventilation can increase arterial PO_2 even when oxygen concentration in inspired air remains unchanged.

True under usual gas-exchange conditions

More fresh alveolar ventilation relative to metabolic CO_2 production → $PaCO_2$ ↓ → PAO_2 ↑ → PaO_2 can rise. FiO_2 remains 21%; it is the renewal of alveolar gas that changes. Improvement does not guarantee restoration of sea-level oxygenation.

Integrated practice 1 | Self-written; not an original exam question

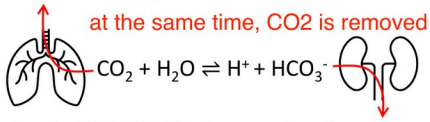
At PB = 380 mmHg, FiO_2 = 0.21, PH_2O = 47 and R = 0.8, $PaCO_2$ falls from 40 to 25 mmHg. Calculate the predicted PAO_2 before/after hyperventilation. Does the inspired oxygen pressure change?
Reasoning: PIO_2 stays 0.21 × (380 − 47) ≈ 70 mmHg. PAO_2 rises from 70 − 40/0.8 ≈ 20 to 70 − 25/0.8 ≈ 39 mmHg. This predicts alveolar pressure, not an identical measured PaO_2. Ventilation improves the result but cannot remove the low-PIO_2 limit.

Source p. 4: “10% per 1000 m” is a rough rule, not an exact linear law; pressure is approximately half sea-level near 5–6 km. Aircraft may cruise at 10–12 km, but a pressurised cabin is not exposed to the full outside pressure [Extension].

03 / Acclimatization: ventilation, kidney and Hb

Lecture PDF p. 5 | Tutorial Q4, Q12 | Original choices checked; BPG mechanism corrected

- Concept: homeostasis will raise the parameter if the parameter drops
  - Sensor-center-effector model
    - The low arterial Po_2 will be detected by peripheral chemoreceptor.
      - Stimulate (hyperventilation/hypoventilation) to increase alveolar Po2



      - But respiratory (acidosis/alkalosis) is resulted.
        - Kidney will (conserve/excrete) bicarbonate ions.

how this interact with the control system?
why we cannot increase the altiture too fast?

  - Lack of oxygen increases glycolysis in erythrocytes
    - (increase/decrease) 2,3-BPG in erythrocytes
      - Oxygen dissociation curve is shifted to the (left/right)
      - Favors oxygen (loading/unloading) – Bohr effect

Original highlighted choices, lung/kidney sketch and prompts (lecture p. 5); reviewed answers below.

Prompt in original order	Checked completion / explanation
Low-PaO_2 response; acid–base change	Peripheral chemoreceptor stimulation → hyperventilation; respiratory alkalosis: $PaCO_2$ ↓ → H^+ ↓ → pH ↑.
Kidney: conserve or excrete?	Excrete more HCO_3^-: loss of base helps compensate for respiratory alkalosis.
RBC 2,3-BPG; curve; tissue effect	Increase via RBC glycolysis; right; unloading. 2,3-bisphosphoglycerate stabilises deoxyHb in the T state.

CORRECTION | BPG right shift is not the Bohr effect

Acute hypocapnia/alkalosis tends to shift the curve left; increasing 2,3-BPG tends to shift it right. The net shift depends on timing and conditions.

Bohr effect means CO_2/H^+ changes alter Hb–O_2 affinity. $PaCO_2$ ↓ → CSF (cerebrospinal fluid) H^+ ↓ → central drive ↓, opposing peripheral hypoxic drive. Renal bicarbonate loss reduces this alkalotic brake over time; it does not increase atmospheric PO_2.

Tutorial Q4 · Original tutorial question

Hyperventilation at high altitude helps improve oxygenation even though it causes respiratory acidosis.

False

The oxygenation benefit can be correct; the primary acute disturbance is respiratory alkalosis. Rapid ascent can outpace acclimatization and increase altitude-illness risk.

Tutorial Q12 · Original tutorial question

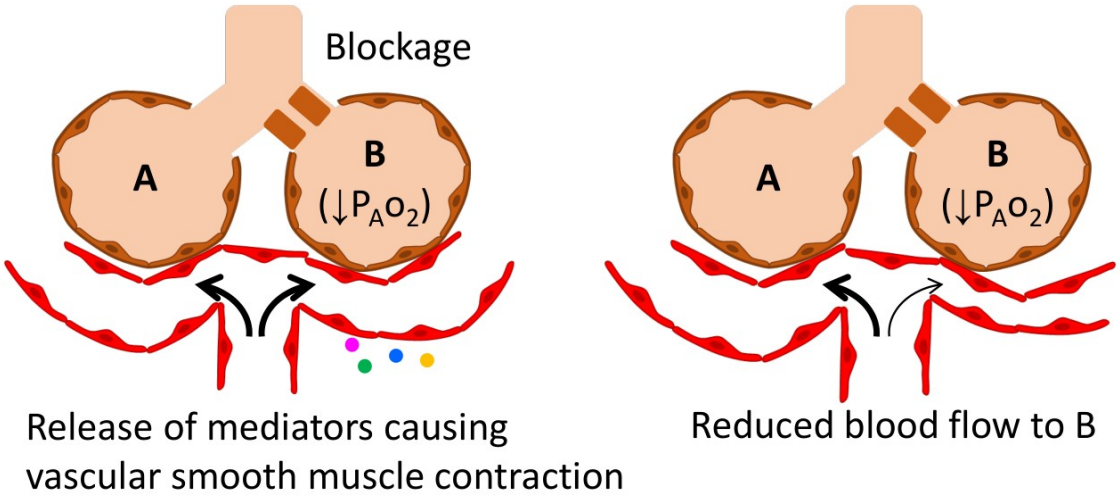
Carbonic anhydrase inhibitor stimulates hyperventilation by promoting bicarbonate ion excretion.

True | Renal mechanism [Extension]

Acetazolamide → renal CA inhibition → HCO_3^- recovery ↓ → urinary base loss ↑ → metabolic acidosis → ventilatory drive ↑. Normal altitude adaptation and the drug both involve bicarbonate loss, but drug-induced base loss is not the original respiratory disturbance.

04 / Hypoxia: pulmonary versus cerebral vessels

Lecture PDF pp. 6–7 | Highlighted directions checked with important conditions

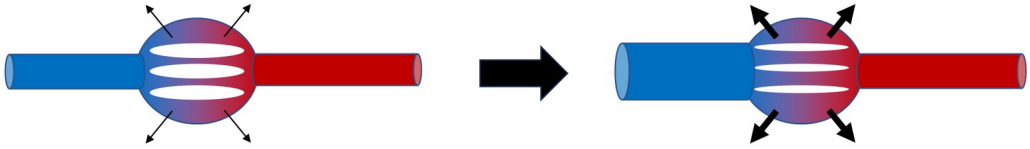


Original local airway-blockage / pulmonary vessel illustration (lecture p. 6).

Checked choices: pulmonary vasoconstriction; blood directed away

Low alveolar PO_2 → pulmonary arteriolar constriction. If only one region is affected, blood is diverted towards better-ventilated units.

At altitude, inspired PO_2 is low across much of the lung even when airways are open. Widespread constriction raises pulmonary vascular resistance and pressure; a local-blockage drawing cannot by itself explain whole-lung hypoxia.



Original capillary fluid-leakage schematic (lecture p. 7); it is a simplified illustration.

Checked choice: cerebral vasodilation for the hypoxic stimulus

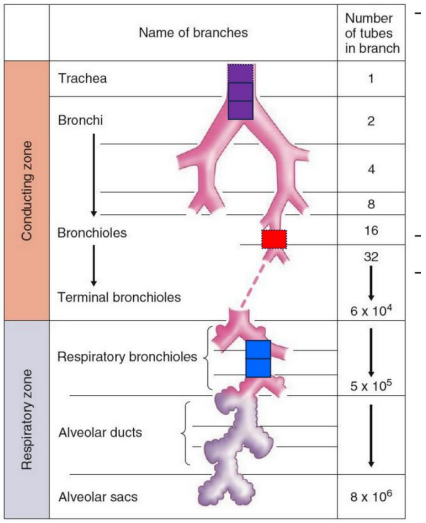
Hypoxia tends to dilate cerebral vessels to support O_2 delivery. Hypocapnia from hyperventilation tends to constrict them; the net response reflects both.

Cerebral edema is not an inevitable result of normal compensatory vasodilation. Severe altitude illness involves disturbed vascular/barrier function as well as pressure effects [Extension].

Distinguish the outcomes [Extension]	Physiological significance
Acute mountain sickness (AMS)	Rapid ascent can exceed adaptation; not synonymous with cerebral edema.
High-altitude pulmonary edema (HAPE)	Hypoxic pulmonary hypertension and uneven microvascular stress can promote lung fluid leakage.
High-altitude cerebral edema (HACE)	Severe brain swelling; more complex than simply “more blood causes edema”.

05 / Dead space: total air versus effective ventilation

Lecture PDF p. 9 | Tutorial Q14



Original conducting/respiratory zone diagram, lecture p. 9.

Term	Definition / significance
Anatomical dead space	Conducting-airway volume: moves air but has no alveolar gas exchange.
Alveolar / physiological dead space [Extension]	Ventilated alveoli with inadequate perfusion / anatomical plus alveolar dead space.
Apparatus dead space [Extension]	Added equipment volume, such as a snorkel, that can increase rebreathing.

Keep volume and rate separate

VT (TV) = tidal volume; VD (DS) = dead-space volume per breath; f = breaths/min.
Minute ventilation $\dot{V}E = VT \times f$. Alveolar ventilation $\dot{V}A = (VT - VD) \times f$ in the simplified model.

Tutorial Q14 · Original tutorial question

Slow deep breathing and rapid shallow breathing provide the same alveolar ventilation if minute ventilation is identical.

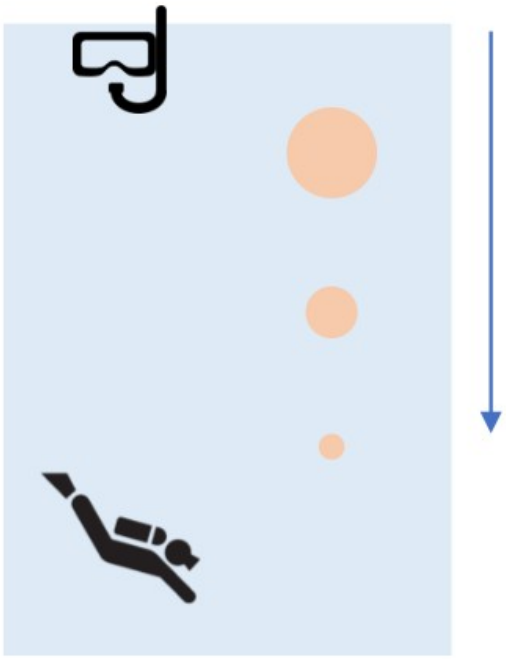
False | Equal minute ventilation can conceal different dead-space losses

With VD = 150 mL: 500 mL × 12/min gives $\dot{V}E$ = 6 L/min and $\dot{V}A$ = 4.2 L/min; 250 mL × 24/min gives the same $\dot{V}E$ but $\dot{V}A$ = 2.4 L/min.

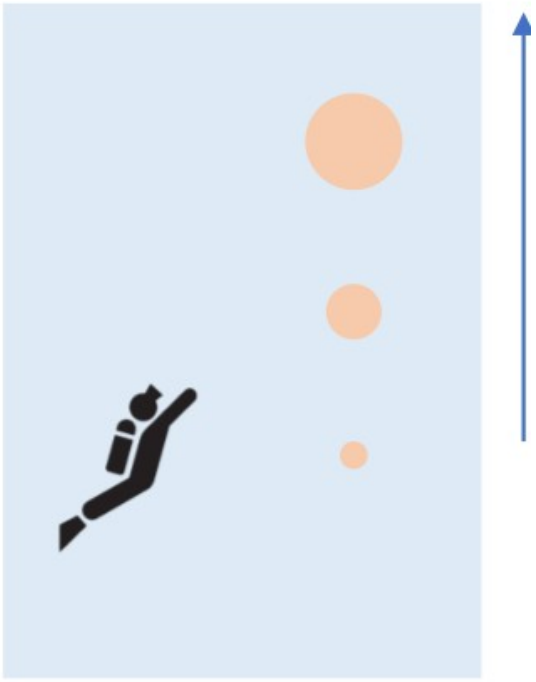
Annotation check: “TV < DS, no ventilation” means little/no fresh convective alveolar ventilation in the idealised model, not no movement of air. Do not calculate a negative physical ventilation rate; real mixing and specialised ventilation modes are not captured by this simple model.

06 / Snorkel versus ambient-pressure gas supply

Lecture PDF p. 10; comparison with p. 11



Original surface-air / depth sketch (p. 10).



Original compressed-air diver sketch (p. 11).

Compare	Surface-air snorkel	Compressed-gas scuba
Inspired gas supply	Communicates with air near surface atmospheric pressure.	Regulator supplies gas approximately at local ambient water pressure [Extension].
Pressure around chest	Water pressure increases with depth; inspired pressure does not track it.	Gas delivery reduces this particular external-pressure mismatch; breathing work is not necessarily normal.
Mechanical consequence	Checked choice: harder to expand the lungs; inspiratory muscles have limited pressure-generating capacity.	Denser gas and equipment can increase respiratory resistance/work [Extension].
Dead space / CO_2	Tube adds apparatus dead space, reducing effective fresh ventilation if VT and f are unchanged.	Depends on equipment and ventilation; compressed gas does not abolish dead-space effects.

What happens if effective ventilation falls?

At unchanged metabolism: CO_2 removal ↓ → $PaCO_2$ ↑; PAO_2 can fall → hypoxemia.
CO_2 stimulates central and peripheral pathways; low PaO_2 stimulates peripheral chemoreceptors. Increased drive is not proof the mechanical/equipment limitation has been overcome.

Pressure and dead space are separate problems

A longer surface-air tube does not solve the rising water pressure around the chest and can add dead space. Example: VT 500 mL, f 12/min; raising effective VD from 150 to 300 mL reduces $\dot{V}A$ from 4.2 to 2.4 L/min, despite unchanged $\dot{V}E$.

07 / Compressed air: pressure, N_2 and O_2

Lecture PDF p. 11 | Tutorial Q13

Fraction can stay constant while partial pressure rises

Increasing absolute ambient pressure → $PN_2 = FN_2 \times P_{total}$ ↑ → equilibrium dissolved N_2 ↑ (Henry). Dissolved concentration = α × partial pressure. Uptake into tissues takes time; composition alone does not determine the dissolved load.

Illustrative seawater depth [Extension]	Absolute pressure	PN_2 at $FN_2 = 0.78$	Dry-gas PO_2 at $FiO_2 = 0.21$
Surface	~1 ATA	0.78 ATA	0.21 ATA
~10 m	~2 ATA	1.56 ATA	0.42 ATA
~30 m	~4 ATA	3.12 ATA	0.84 ATA

ATA = atmosphere absolute; use absolute pressure, not gauge pressure. These are dry-gas illustrative values, not a dive plan or identical arterial measurements.

Tutorial Q13 · Original tutorial question

A diver breathing 78% N2 and 21% O2 will have blood dissolved nitrogen level remains unchanged because the fraction of nitrogen is unchanged.

False | Total pressure is the missing variable

FN_2 unchanged does not mean PN_2 unchanged. Higher total pressure increases PN_2 and, after equilibration, dissolved nitrogen.

CORRECTION | More dissolved gas does not mean α has increased

“Normally poorly soluble has now become soluble” is misleading. At fixed temperature and medium, the solubility coefficient remains approximately fixed; concentration increases because partial pressure increases.

- O_2 partial pressure and dissolved O_2 also rise; this is not unique to nitrogen. Hb-bound O_2 has limited extra capacity when Hb is already nearly saturated.
- High nitrogen partial pressure can impair CNS function: nitrogen narcosis, compared with intoxication in the slide. This is distinct from bubble formation during decompression.
- [Extension] High oxygen partial pressure has separate toxicity effects; it is not the mechanism of nitrogen narcosis.

08 / Ascent: dissolved-gas bubbles versus trapped-gas expansion

Lecture PDF p. 11 | Mechanisms separated explicitly

Condition	Trigger / gas source	Relevant mechanism
Nitrogen narcosis	High nitrogen partial pressure at depth; no bubbles required.	High-pressure gas effect on CNS function.
Decompression sickness (DCS)	Pressure falls while tissues contain excess dissolved inert gas; gas can form bubbles in tissue/blood.	Henry-related supersaturation and gas elimination kinetics.
Pulmonary barotrauma / arterial gas embolism (AGE)	Trapped alveolar gas expands as external pressure falls; lung injury can permit gas to enter pulmonary veins and systemic arteries.	Boyle's law: expansion of gas already present in a gas space.

Two ascent pathways

DCS: pressure ↓ → equilibrium dissolved-gas concentration ↓ → excess N_2 may come out of solution.
AGE from lung overexpansion: pressure ↓ → trapped alveolar gas volume ↑ → possible tissue rupture and entry of gas into arteries.
Ascent does not automatically rupture normal lungs: gas trapping / inability to vent expansion is an important condition.

Integrated practice 2 | Self-written; not an original exam question

A sealed 1 L gas pocket at 2 ATA is brought to 1 ATA at constant temperature. What volume does Boyle's law predict? Why does that calculation not explain nitrogen decompression sickness?
Reasoning: $P_1V_1 = P_2V_2$ predicts 2 L if the pocket can freely expand; a constrained space may instead be injured. The calculation concerns trapped gas, not dissolved nitrogen. DCS concerns supersaturation/outgassing; AGE can arise from alveolar gas entering arteries. Both can follow ascent, but they need not share the same gas origin or mechanism.

[Extension] Decompression illness is an umbrella term that can include DCS and AGE. Their mechanisms should not be merged into "all bubbles are nitrogen leaving solution".

09 / Source audit: what to correct or qualify

Lecture PDF pp. 4–12 | Original summary retained for comparison

- ✓ Exercise doesn't cause significant changes in blood gases and pH due to the compensatory increase in ventilation (homeostasis)
- ✓ Ascend to high altitude too quickly causes acute mountain sickness because low atmospheric pressure reduces alveolar Po_2 (Boyle's law) which lead to hypoxia.
- ✓ Hypoxia-induced hyperventilation at high altitude induces respiratory alkalosis which stimulates hypoventilation.
- ✓ Ascend from deep sea too quickly causes decompression illness because rapid reduction in pressure allows dissolved nitrogen to form bubbles (Henry's law)

Original highlighted summary, lecture p. 12. Its claims are reviewed below; the Boyle attribution needs correction.

Source statement / selection	Reviewed interpretation
p. 5: hyperventilation / alkalosis / excrete	All three highlighted choices are correct. Remaining completions: increase BPG → right shift → unloading.
p. 5: calls BPG unloading "Bohr effect"	CORRECTION: BPG stabilises the T state; Bohr refers to CO_2/H^+ effects. Acute hypocapnia and later BPG changes may oppose each other.
pp. 6–7: pulmonary constriction / away; cerebral dilation	Selected directions are correct for the stated local/hypoxic effects. Global altitude hypoxia, hypocapnia and edema need the qualifications on p. 4 of these notes.
p. 9: TV < DS, no ventilation	A limit of a simple fresh-gas model, not absence of all airflow or a physically negative ventilation rate.
p. 11: high pressure means high "solubility"	CORRECTION: dissolved concentration rises; α need not change.
p. 12: altitude hypoxia attributed to Boyle	CORRECTION: falling barometric pressure lowers inspired PO_2 through Dalton; Henry relates the dissolved component. Boyle explains volume changes.
p. 12: exercise does not change gases or pH	Often approximately true for moderate exercise with compensation; not universal at high intensity or in disease.
p. 12: alkalosis stimulates hypoventilation	An opposing inhibitory drive, not a guarantee that overall ventilation becomes lower than baseline at altitude.

Time matters: increased ventilation begins promptly; renal/CSF adaptation and RBC BPG changes develop over time. [Extension] Increased erythropoietin/RBC mass can improve carrying capacity over longer acclimatization, but is not an immediate correction of inspired PO_2. There are two self-written integrated problems; all other numbered questions come from the supplied tutorial.